

Chap 5. The Discrete-time Fourier Transform

- Discrete-Time Aperiodic Signals
- Properties of the Discrete-Time Fourier Transform
- Convolution Property
- Duality

- Continuous-Time Fourier Series (CTFS)
- Discrete-Time Fourier Series (DTFS)
- Continuous-Time Fourier Transform (CTFT)
- Discrete-Time Fourier Transform (DTFT)
- Discrete Fourier Transform (DFT)
- Fast Fourier Transform (FFT)

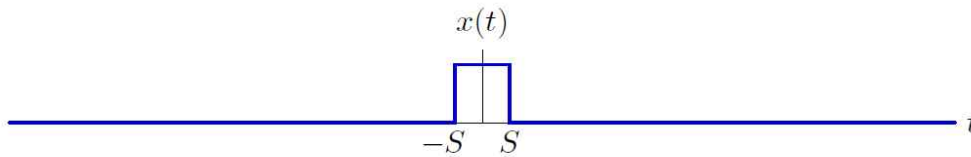
§5.1 Representation of Aperiodic Signals:

The Discrete-Time Fourier Transform (DTFT)

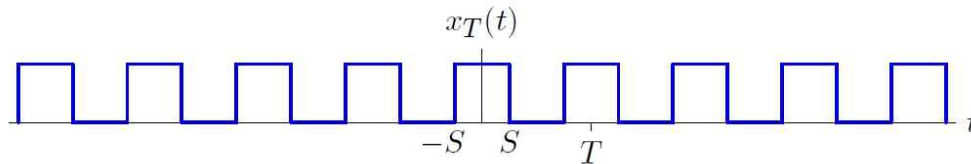
※ CTFS / CTFT (Chapter 4)

An aperiodic signal can be thought of as periodic with infinite period.

Let $x(t)$ represent an aperiodic signal.

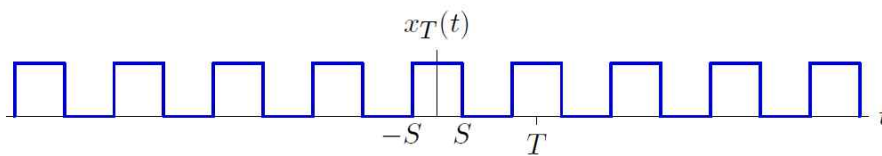


“Periodic extension”:
$$x_T(t) = \sum_{k=-\infty}^{\infty} x(t + kT)$$

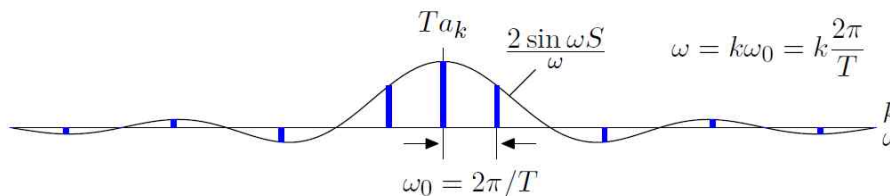


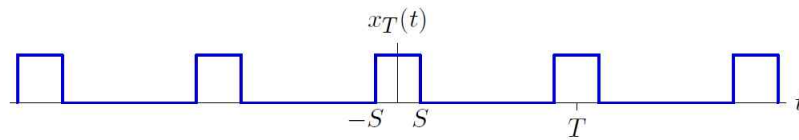
Then $x(t) = \lim_{T \rightarrow \infty} x_T(t)$.

Represent $x_T(t)$ by its Fourier series.

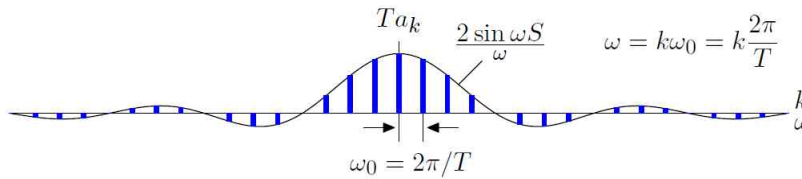


$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x_T(t) e^{-j\frac{2\pi}{T}kt} dt = \frac{1}{T} \int_{-S}^S e^{-j\frac{2\pi}{T}kt} dt = \frac{\sin \frac{2\pi kS}{T}}{\pi k} = \frac{2 \sin \omega S}{T \omega}$$

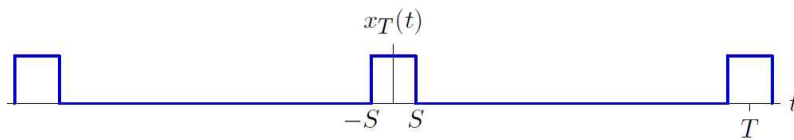


Period (T) doubles, pulse width (τ) fixed

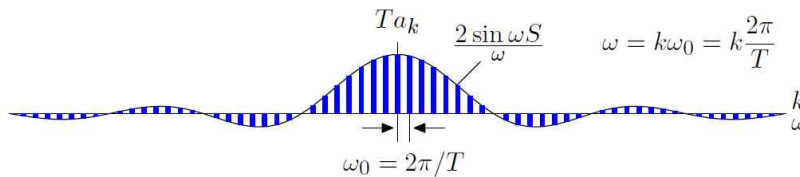
$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x_T(t) e^{-j\frac{2\pi}{T}kt} dt = \frac{1}{T} \int_{-S}^S e^{-j\frac{2\pi}{T}kt} dt = \frac{\sin \frac{2\pi kS}{T}}{\pi k} = \frac{2}{T} \frac{\sin \omega S}{\omega}$$



T bigger and bigger, ω_0 (or $f_0 = \frac{1}{T}$) smaller and smaller)

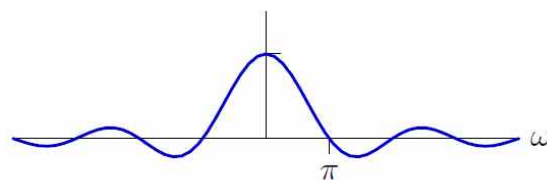
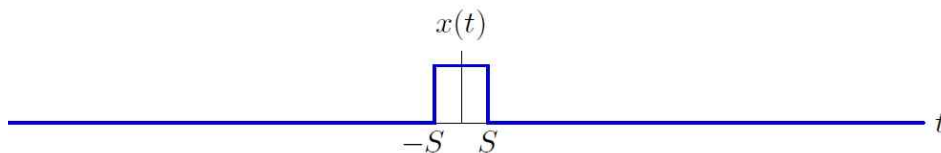


$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x_T(t) e^{-j\frac{2\pi}{T}kt} dt = \frac{1}{T} \int_{-S}^S e^{-j\frac{2\pi}{T}kt} dt = \frac{\sin \frac{2\pi kS}{T}}{\pi k} = \frac{2}{T} \frac{\sin \omega S}{\omega}$$



$$\lim_{T \rightarrow \infty} T a_k = \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} x(t) e^{-j\omega t} dt = \frac{2}{\omega} \sin \omega S = E(\omega)$$

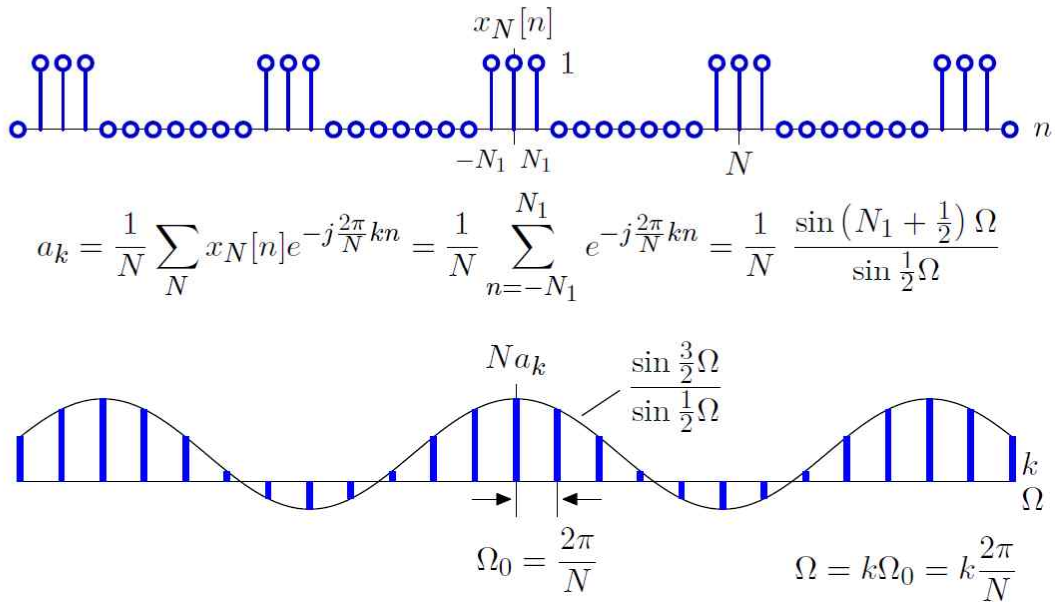
$T \rightarrow \infty$ (nonperiodic function), ω_0 (or $f_0 = \frac{1}{T}$) $\rightarrow 0$



$$X(j\omega) = \frac{2 \sin(\omega S)}{\omega}$$

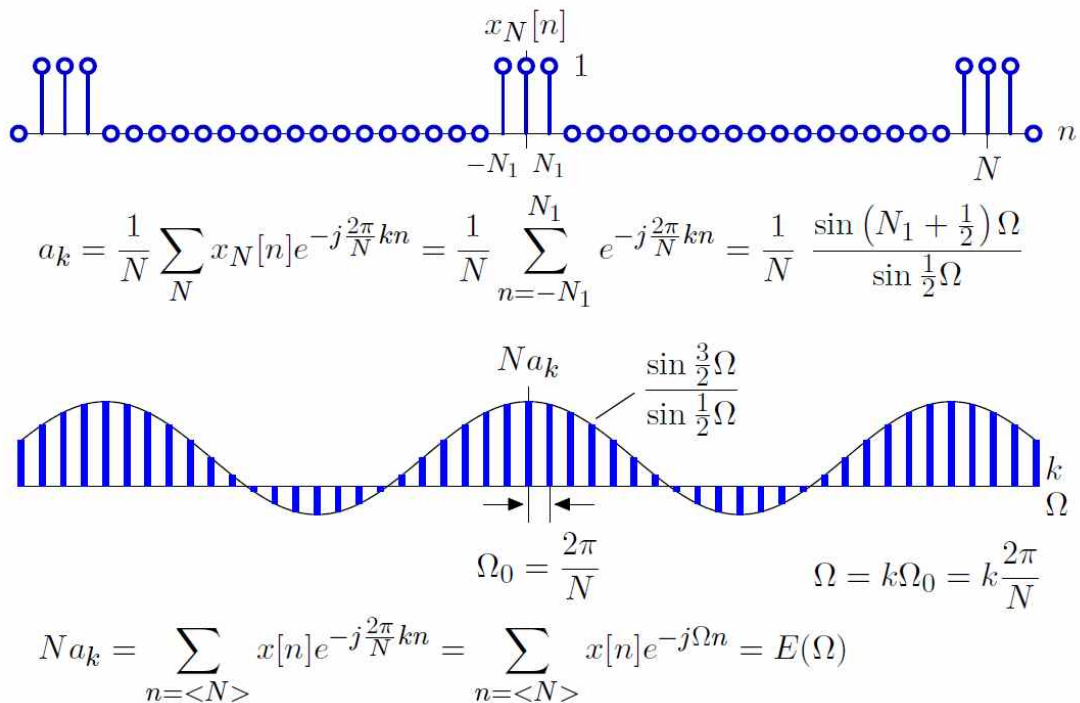
- Period gets bigger (N gets bigger)

Doubling period doubles # of harmonics in given frequency interval.

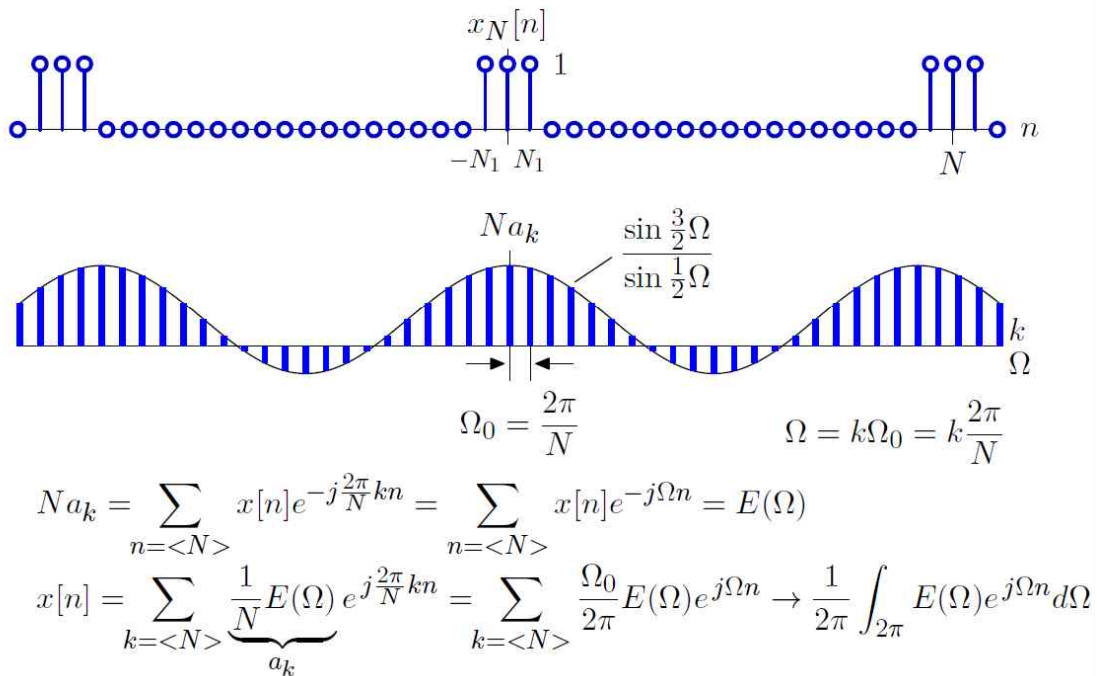


- period $\rightarrow \infty$ ($N \rightarrow \infty$)

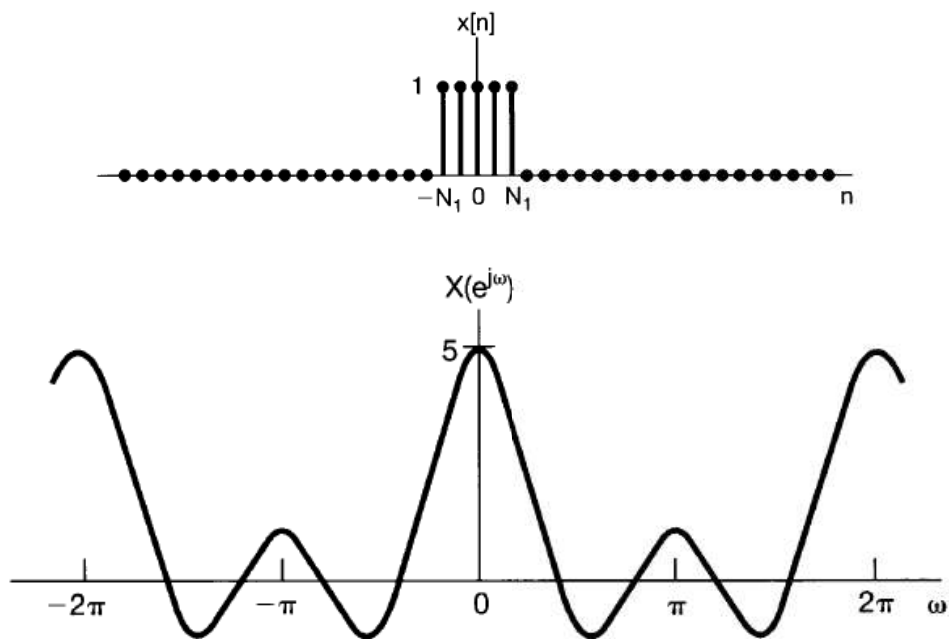
As $N \rightarrow \infty$, discrete harmonic amplitudes $\rightarrow$ a continuum $E(\Omega)$.

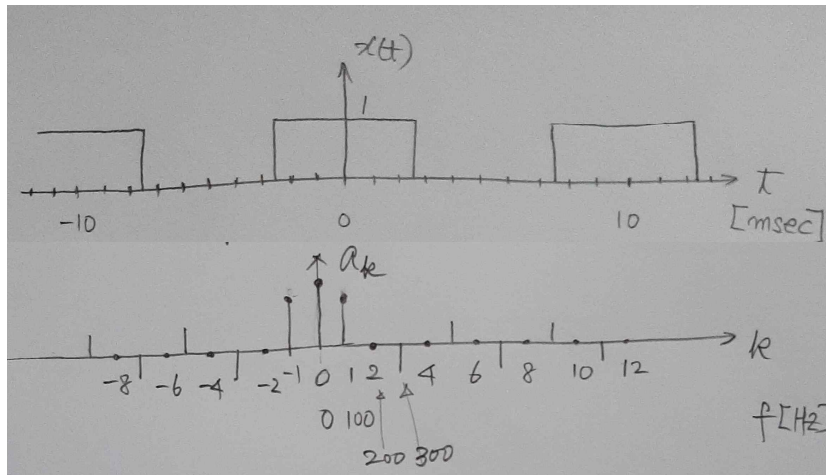
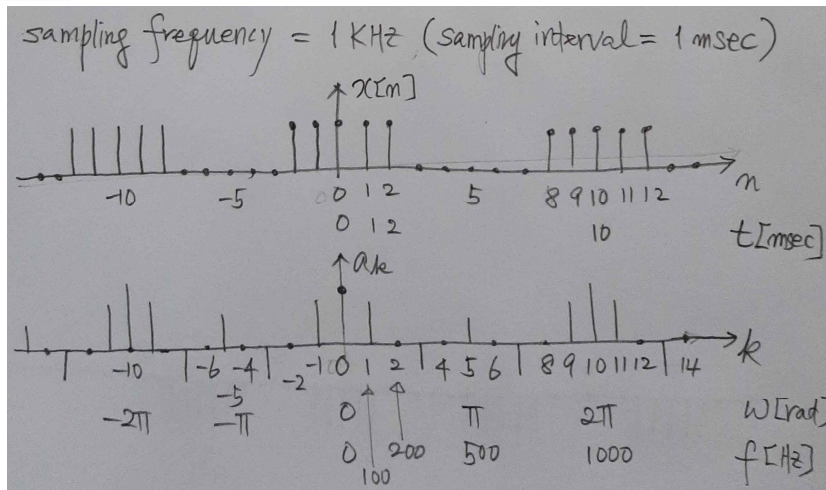
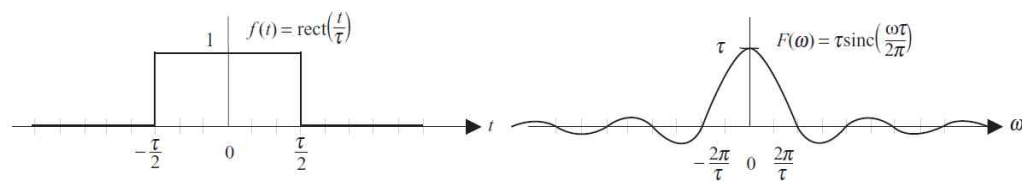
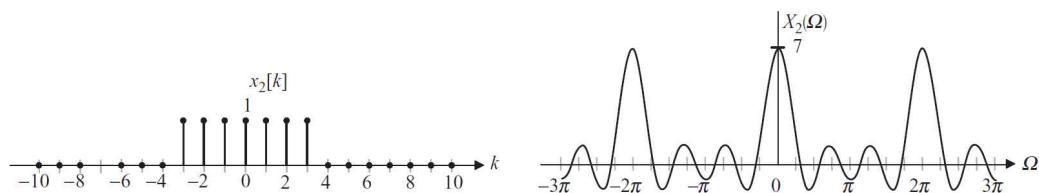


As $N \rightarrow \infty$, synthesis sum $\rightarrow$ integral.



- Finally period $\rightarrow \infty$ ($N \rightarrow \infty$)



CTFSDTFSCTFTDTFT

§5.1.1 Development of the Discrete-Time Fourier Transform

- Finite duration signal $x[n]$ / Periodic signal $\tilde{x}[n]$

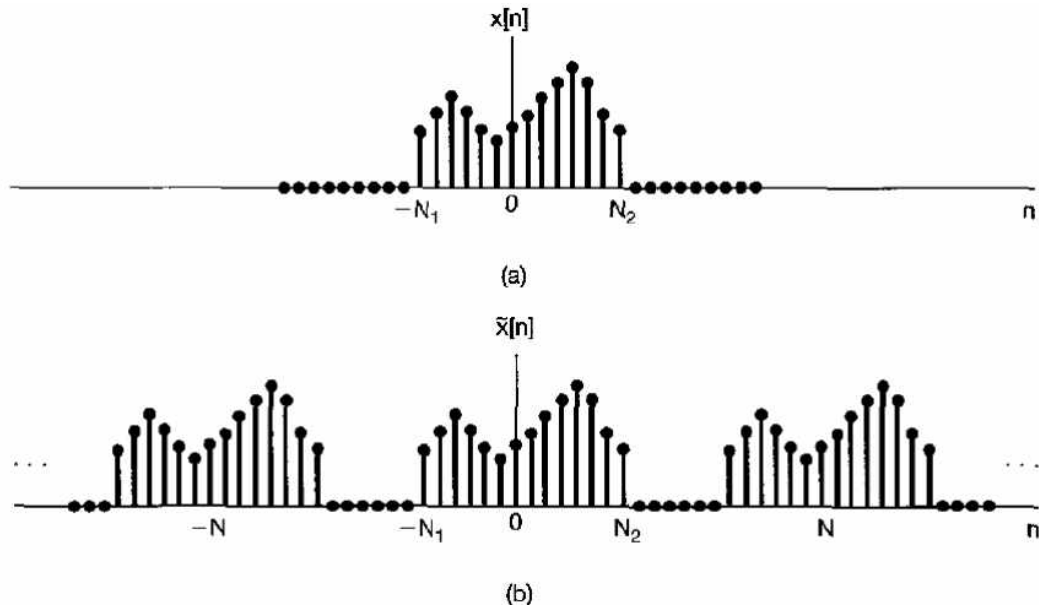


Fig. 5.1 (a) Finite-duration signal $x[n]$, (b) periodic signal $\tilde{x}[n]$

Fourier series representation of $\tilde{x}[n]$

$$\tilde{x}[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} \tilde{x}[n] e^{-jk(2\pi/N)n}$$

Over a period $x[n] = \tilde{x}[n]$

$$a_k = \frac{1}{N} \sum_{n=-N_1}^{N_2} x[n] e^{-jk(2\pi/N)n} = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk(2\pi/N)n}$$

Define the function $X(e^{j\omega})$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-jk(2\pi/N)n}$$

Coefficient a_k are proportional to samples of $X(e^{j\omega})$

$$a_k = \frac{1}{N} X(e^{jk\omega_0}) \quad \text{where } \omega_0 = 2\pi/N$$

Signal $\tilde{x}[n]$

$$\tilde{x}[n] = \sum_{k=\langle N \rangle} \frac{1}{N} X(e^{jk\omega_0}) e^{jk(2\pi/N)n} = \frac{1}{2\pi} \sum_{k=\langle N \rangle} X(e^{jk\omega_0}) e^{jk(2\pi/N)n\omega_0}$$

As $N \rightarrow \infty$, $\omega_0 \rightarrow 0$, $x[n] = \tilde{x}[n]$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

Discrete-Time Fourier Transform (DTFT) Pair

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \quad (5.8)$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} \quad (5.9)$$

* $X(e^{j\omega})$ is the spectrum of a aperiodic signal $x[n]$.

- DTFS $\Rightarrow$ DTFT

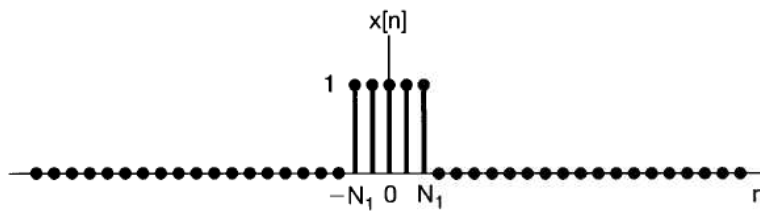
Discrete-time periodic signal $\tilde{x}[n]$		Discrete-time aperiodic signal $x[n]$
Discrete-Time Fourier Series a_k (DTFS)		Discrete-Time Fourier Transform $X(e^{j\omega})$ (DTFT)
$\tilde{x}[n] = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 n}$ $= \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/N)n}$	$N \rightarrow \infty$ $\omega_0 \rightarrow 0$	$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$
$a_k = \frac{1}{N} \sum_{k=\langle N \rangle} \tilde{x}(t) e^{-jk\omega_0 n}$ $= \frac{1}{N} \sum_{k=\langle N \rangle} \tilde{x}(t) e^{-jk(2\pi/N)n}$		$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n}$

§5.1.2 Examples of Discrete-Time Fourier Transforms

(Example 5.3) Consider the rectangular pulse

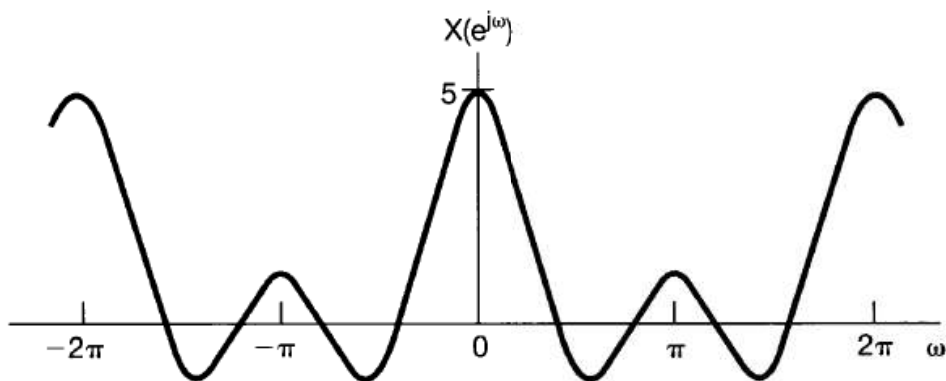
$$x[n] = \begin{cases} 1, & |n| \leq N_1 \\ 0, & |n| > N_1 \end{cases}$$

Example for $N_1 = 2$



Fourier Transform

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-N_1}^{+N_1} x[n] e^{-j\omega n} = e^{-j\omega(-N_1)} + e^{-j\omega(-(N_1-1))} + \dots + e^{-j\omega(N_1)} \\ &= \frac{e^{-j\omega(-N_1)} \times (1 - e^{-j\omega(2N_1+1)})}{1 - e^{-j\omega}} = \frac{e^{j\omega(N_1+1/2)} - e^{-j\omega(N_1+1/2)}}{e^{j\omega/2} - e^{-j\omega/2}} \\ &= \frac{\sin(\omega(N_1+0.5))}{\sin(\omega/2)} \end{aligned}$$



§5.2 The Fourier Transform for Periodic Signals

(Example 5.6)

Periodic impulse train

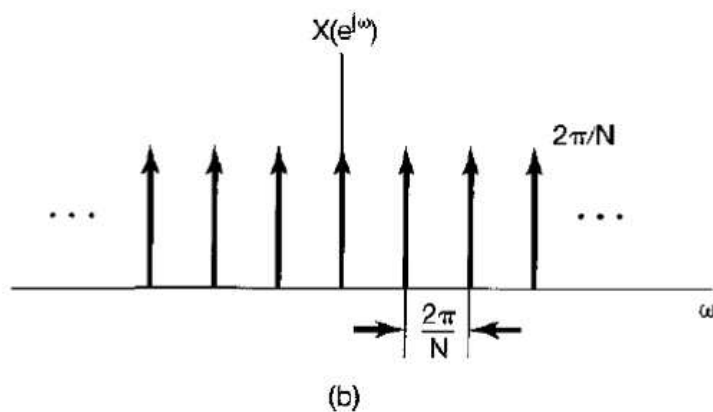
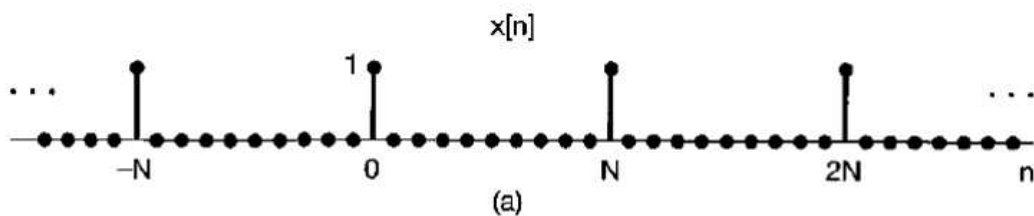
$$x[n] = \sum_{k=-\infty}^{+\infty} a_k \delta[n - kN]$$

Fourier series

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n} = \frac{1}{N}$$

Fourier transform

$$X(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{N}\right)$$



§5.3 Properties of the DTFT

- Notation

$$X(e^{j\omega}) = F\{x[n]\}$$

$$x[n] = F^{-1}\{X(e^{j\omega})\}$$

$$x[n] \xleftrightarrow{F} X(e^{j\omega})$$

§5.3.1 Periodicity of the Discrete-Time Fourier Transform

$$X(e^{j(\omega+2\pi)}) = X(e^{j\omega}) \quad (5.28)$$

§5.3.2 Linearity

- If $x_1[n] \xleftrightarrow{F} X_1(e^{j\omega})$, $x_2[n] \xleftrightarrow{F} X_2(e^{j\omega})$

$$ax_1[n] + bx_2[n] \xleftrightarrow{F} aX_1(e^{j\omega}) + bX_2(e^{j\omega}) \quad (5.29)$$

§5.3.3 Time Shifting & Frequency Shifting

$$x[n - n_0] \xleftrightarrow{F} e^{-j\omega n_0} X(e^{j\omega}) \quad (5.30)$$

$$e^{-j\omega_0 n} x[n - n_0] \xleftrightarrow{F} X(e^{j(\omega - \omega_0)}) \quad (5.31)$$

§5.3.4 Conjugation and Conjugate Symmetry

$$x^*[n] \xleftrightarrow{F} X^*(e^{-j\omega}) \quad (5.35)$$

- Conjugate Symmetry

$$\text{If } x[n] \text{ is real, then } X(e^{j\omega}) = X^*(e^{-j\omega}). \quad (5.36)$$

§5.3.5 Differencing and Accumulation

$$x[n] - x[n-1] \xleftrightarrow{F} (1 - e^{-j\omega})X(e^{j\omega}) \quad (5.37)$$

§5.3.6 Time Reversal

$$x[-n] \xleftrightarrow{F} X(e^{-j\omega}) \quad (5.42)$$

$$(5.45)$$

§5.4 The Convolution Property

$$y[n] = x[n] \star h[n] \leftrightarrow Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega}) \quad (5.48)$$

(Example 5.11) time delay

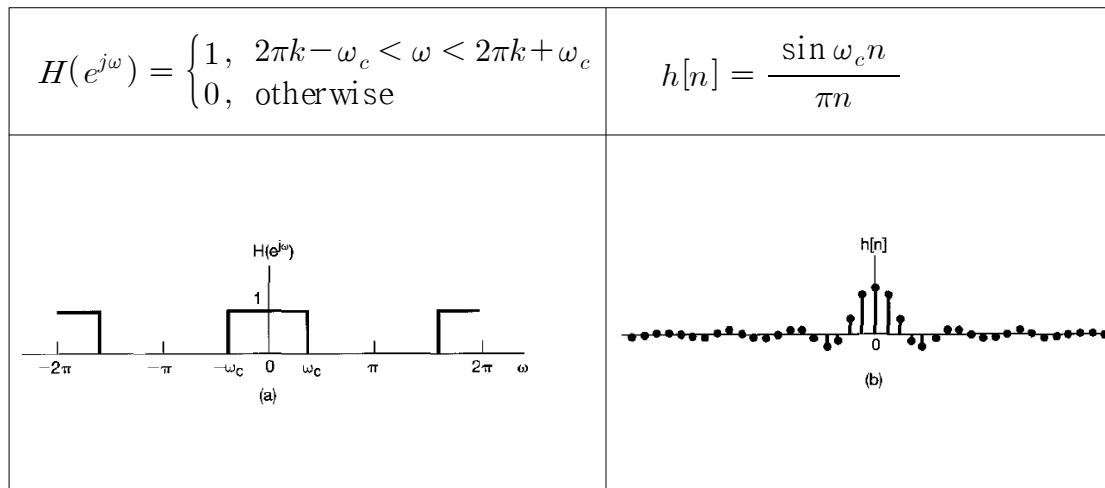
$$h[n] = \delta[n - n_0]$$

$$H(e^{j\omega}) = e^{-j\omega n_0}$$

$$Y(e^{j\omega}) = H(e^{j\omega}) X(e^{j\omega}) = e^{-j\omega n_0} X(e^{j\omega})$$

$$y[n] = x[n - n_0]$$

(Example 5.12) low-pass filtering (ideal low-pass filter [LPF])



$$h[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega = \frac{\sin \omega_c n}{\pi n}$$

§5.5 The Multiplication Property

$$x_1[n] \leftrightarrow X_1(e^{-j\omega})$$

$$x_2[n] \leftrightarrow X_2(e^{-j\omega})$$

$$y[n] = x_1[n] x_2[n] \text{ Then}$$

$$Y(e^{j\omega}) = \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\theta}) X_2(e^{j(\omega-\theta)}) d\theta \quad (5.63)$$

§5.6 Tables of DTFT Pairs

TABLE 5.1 PROPERTIES OF THE DISCRETE-TIME FOURIER TRANSFORM

Section	Property	Aperiodic Signal	Fourier Transform
		$x[n]$	$X(e^{j\omega})$ periodic with
		$y[n]$	$Y(e^{j\omega})$ period 2π
5.3.2	Linearity	$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$
5.3.3	Time Shifting	$x[n - n_0]$	$e^{-j\omega n_0} X(e^{j\omega})$
5.3.3	Frequency Shifting	$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$
5.3.4	Conjugation	$x^*[n]$	$X^*(e^{-j\omega})$
5.3.6	Time Reversal	$x[-n]$	$X(e^{-j\omega})$
5.3.7	Time Expansion	$x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n = \text{multiple of } k \\ 0, & \text{if } n \neq \text{multiple of } k \end{cases}$	$X(e^{jk\omega})$
5.4	Convolution	$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$
5.5	Multiplication	$x[n]y[n]$	$\frac{1}{2\pi} \int_{2\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)}) d\theta$
5.3.5	Differencing in Time	$x[n] - x[n-1]$	$(1 - e^{-j\omega})X(e^{j\omega})$
5.3.5	Accumulation	$\sum_{k=-\infty}^n x[k]$	$\frac{1}{1 - e^{-j\omega}} X(e^{j\omega})$ $+ \pi X(e^{j0}) \sum_{k=-\infty}^{+\infty} \delta(\omega - 2\pi k)$
5.3.8	Differentiation in Frequency	$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$
5.3.4	Conjugate Symmetry for Real Signals	$x[n]$ real	$\begin{cases} X(e^{j\omega}) = X^*(e^{-j\omega}) \\ \Re\{X(e^{j\omega})\} = \Re\{X(e^{-j\omega})\} \\ \Im\{X(e^{j\omega})\} = -\Im\{X(e^{-j\omega})\} \\ X(e^{j\omega}) = X(e^{-j\omega}) \\ \angle X(e^{j\omega}) = -\angle X(e^{-j\omega}) \end{cases}$
5.3.4	Symmetry for Real, Even Signals	$x[n]$ real and even	$X(e^{j\omega})$ real and even
5.3.4	Symmetry for Real, Odd Signals	$x[n]$ real and odd	$X(e^{j\omega})$ purely imaginary and odd
5.3.4	Even-odd Decomposition of Real Signals	$x_e[n] = \mathcal{E}\{x[n]\}$ [$x[n]$ real] $x_o[n] = \mathcal{O}\{x[n]\}$ [$x[n]$ real]	$\Re\{X(e^{j\omega})\}$ $j\Im\{X(e^{j\omega})\}$
5.3.9	Parseval's Relation for Aperiodic Signals		
		$\sum_{n=-\infty}^{+\infty} x[n] ^2 = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) ^2 d\omega$	

TABLE 5.2 BASIC DISCRETE-TIME FOURIER TRANSFORM PAIRS

Signal	Fourier Transform	Fourier Series Coefficients (if periodic)
$\sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/N)n}$	$2\pi \sum_{k=-\infty}^{\infty} a_k \delta\left(\omega - \frac{2\pi k}{N}\right)$	a_k
$e^{j\omega_0 n}$	$2\pi \sum_{l=-\infty}^{\infty} \delta(\omega - \omega_0 - 2\pi l)$	(a) $\omega_0 = \frac{2\pi m}{N}$ $a_k = \begin{cases} 1, & k = m, m \pm N, m \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic
$\cos \omega_0 n$	$\pi \sum_{l=-\infty}^{\infty} \{\delta(\omega - \omega_0 - 2\pi l) + \delta(\omega + \omega_0 - 2\pi l)\}$	(a) $\omega_0 = \frac{2\pi m}{N}$ $a_k = \begin{cases} \frac{1}{2}, & k = \pm m, \pm m \pm N, \pm m \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic
$\sin \omega_0 n$	$\frac{\pi}{j} \sum_{l=-\infty}^{\infty} \{\delta(\omega - \omega_0 - 2\pi l) - \delta(\omega + \omega_0 - 2\pi l)\}$	(a) $\omega_0 = \frac{2\pi r}{N}$ $a_k = \begin{cases} \frac{1}{2j}, & k = r, r \pm N, r \pm 2N, \dots \\ -\frac{1}{2j}, & k = -r, -r \pm N, -r \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational $\Rightarrow$ The signal is aperiodic
$x[n] = 1$	$2\pi \sum_{l=-\infty}^{\infty} \delta(\omega - 2\pi l)$	$a_k = \begin{cases} 1, & k = 0, \pm N, \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases}$
Periodic square wave $x[n] = \begin{cases} 1, & n \leq N_1 \\ 0, & N_1 < n \leq N/2 \end{cases}$ and $x[n+N] = x[n]$	$2\pi \sum_{k=-\infty}^{\infty} a_k \delta\left(\omega - \frac{2\pi k}{N}\right)$	$a_k = \frac{\sin[(2\pi k/N)(N_1 + \frac{1}{2})]}{N \sin[2\pi k/2N]}, \quad k \neq 0, \pm N, \pm 2N, \dots$ $a_k = \frac{2N_1 + 1}{N}, \quad k = 0, \pm N, \pm 2N, \dots$
$\sum_{k=-\infty}^{\infty} \delta[n - kN]$	$\frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{N}\right)$	$a_k = \frac{1}{N}$ for all k
$a^n u[n], \quad a < 1$	$\frac{1}{1 - ae^{-j\omega}}$	—
$x[n] = \begin{cases} 1, & n \leq N_1 \\ 0, & n > N_1 \end{cases}$	$\frac{\sin[\omega(N_1 + \frac{1}{2})]}{\sin(\omega/2)}$	—
$\frac{\sin Wn}{\pi n} = \frac{W}{\pi} \text{sinc}\left(\frac{Wn}{\pi}\right)$ $0 < W < \pi$	$X(\omega) = \begin{cases} 1, & 0 \leq \omega \leq W \\ 0, & W < \omega \leq \pi \end{cases}$ $X(\omega)$ periodic with period 2π	—
$\delta[n]$	1	—
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi \delta(\omega - 2\pi k)$	—
$\delta[n - n_0]$	$e^{-j\omega n_0}$	—
$(n+1)a^n u[n], \quad a < 1$	$\frac{1}{(1 - ae^{-j\omega})^2}$	—
$\frac{(n+r-1)!}{n!(r-1)!} a^n u[n], \quad a < 1$	$\frac{1}{(1 - ae^{-j\omega})^r}$	—

§5.7 Duality

§5.7.2 Duality between the DTFT and CTFS

- DTFT pair

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-jk\omega n} \quad (\text{Period: } 2\pi)$$

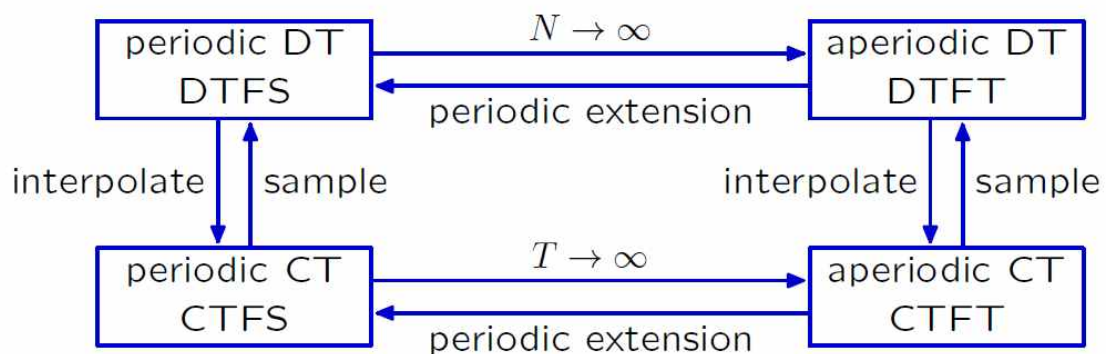
- CTFS pair

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{-jk\omega_0 t} \quad (\text{Period: } T)$$

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$$

TABLE 5.3 SUMMARY OF FOURIER SERIES AND TRANSFORM EXPRESSIONS

	Continuous time		Discrete time	
	Time domain	Frequency domain	Time domain	Frequency domain
Fourier Series	$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$ continuous time periodic in time	$a_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$ discrete frequency aperiodic in frequency	$x[n] = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/N)n}$ discrete time periodic in time	$a_k = \frac{1}{N} \sum_{n=-\infty}^{+\infty} x[n] e^{-jk(2\pi/N)n}$ discrete frequency periodic in frequency
Fourier Transform	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$ continuous time aperiodic in time	$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$ continuous frequency aperiodic in frequency	$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$ discrete time aperiodic in time	$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n}$ continuous frequency periodic in frequency



Summary

● Continuous-Time Fourier Series (CTFS) Pair

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t} \quad (3.38)$$

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_T x(t) e^{-jk(2\pi/T)t} dt \quad (3.39)$$

● Discrete-Time Fourier Series (DTFS) pair

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n} \quad (3.94)$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n} \quad (3.95)$$

● Continuous-Time Fourier Transform (CTFT) Pair

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \quad (4.8)$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (4.9)$$

● Discrete-Time Fourier Transform (DTFT) Pair

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \quad (5.8)$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} \quad (5.9)$$